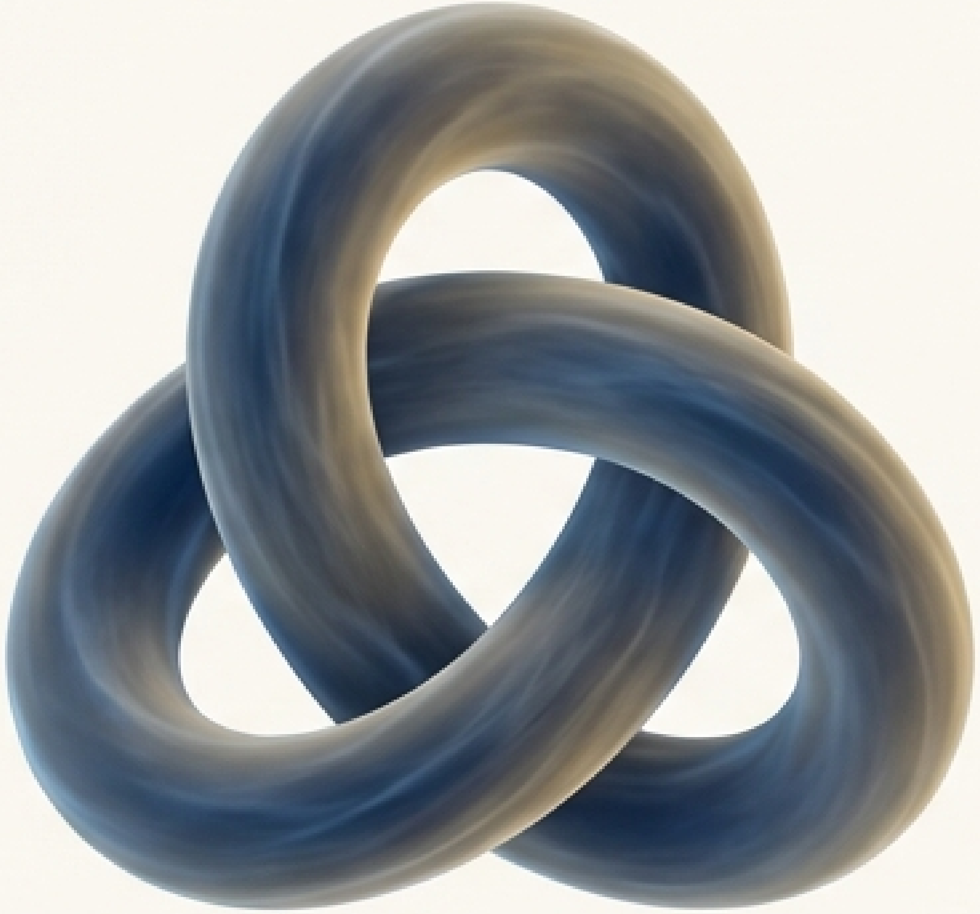
Kelvin Mode Suppression in Atomic Orbitals

Solving a Thermodynamic Catastrophe in Vortex-Filament Models



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Based on the paper: Kelvin Mode Suppression in Atomic Orbitals: A Vortex-Filament Gap.

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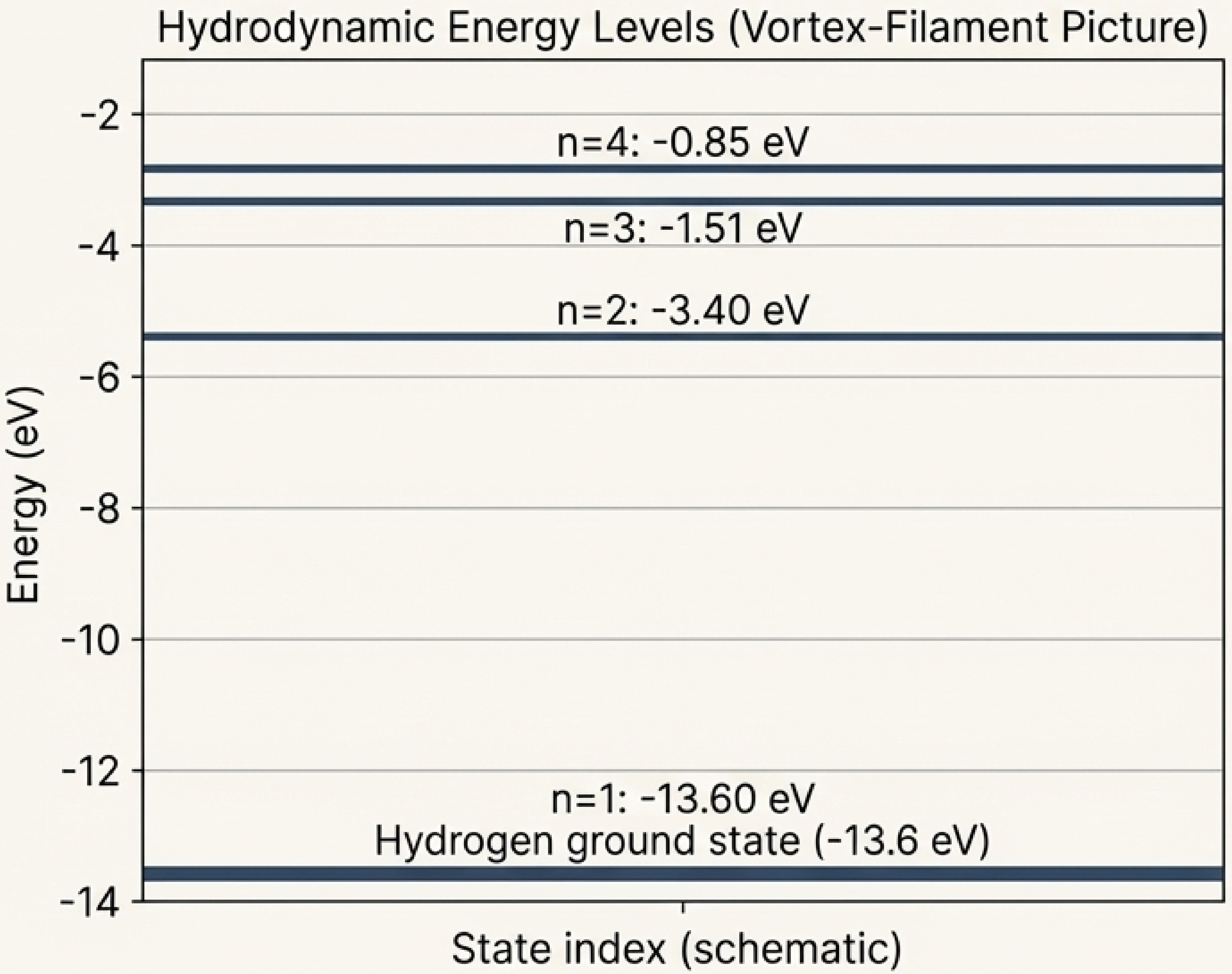
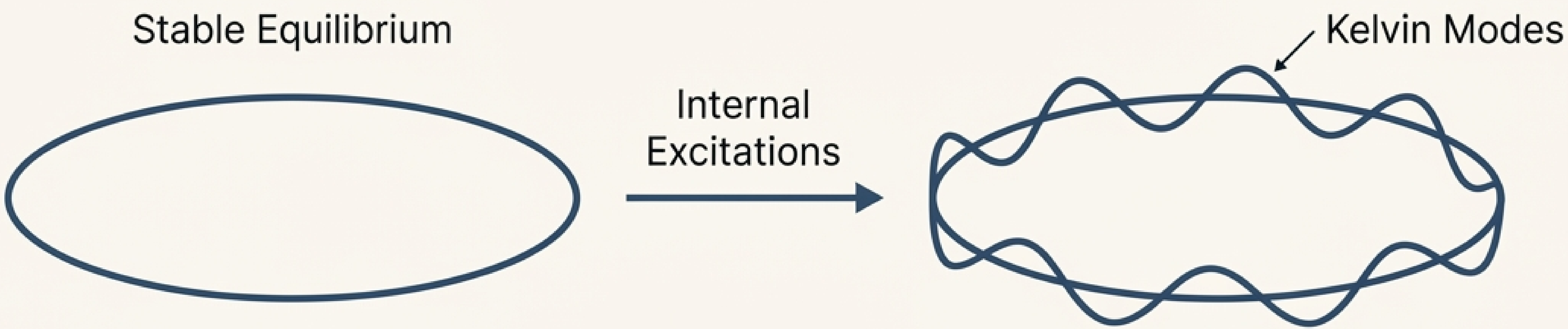
A Classical Picture for a Quantum World



The model's equilibrium states match the known quantized energy levels of the hydrogen atom.

Hydrodynamic models can represent electrons as knotted vortex filaments, correctly reproducing the hydrogen atom's energy spectrum as stable equilibrium states.

The Hidden Flaw: A Filament's Internal Life



These filaments are not rigid structures. They support internal, helical waves known as Kelvin modes.

The key intuition is that longer, higher-energy orbitals are 'softer' and their waves are more easily excited.

Kelvin Wave Frequency Scaling for a Closed Loop:

$$\omega_m \propto \frac{\Gamma m^2}{L^2}$$

A Prediction at War with Reality

Model Prediction

Thermal energy from Kelvin waves would cause massive, measurable shifts in atomic energy levels.

The predicted thermal correction coefficient is

$$a_{\text{naive},n} \sim 10^{-39}\ \text{J/K}^2.$$

Experimental Reality

Atomic spectra are among the most stable and precise measurements in science.

Spectroscopic limits require the thermal correction coefficient to be

$$a_n \lesssim 10^{-62}\ \text{J/K}^2.$$

The Discrepancy:

>20 Orders of Magnitude

Without a suppression mechanism, the vortex-filament model is catastrophically wrong.

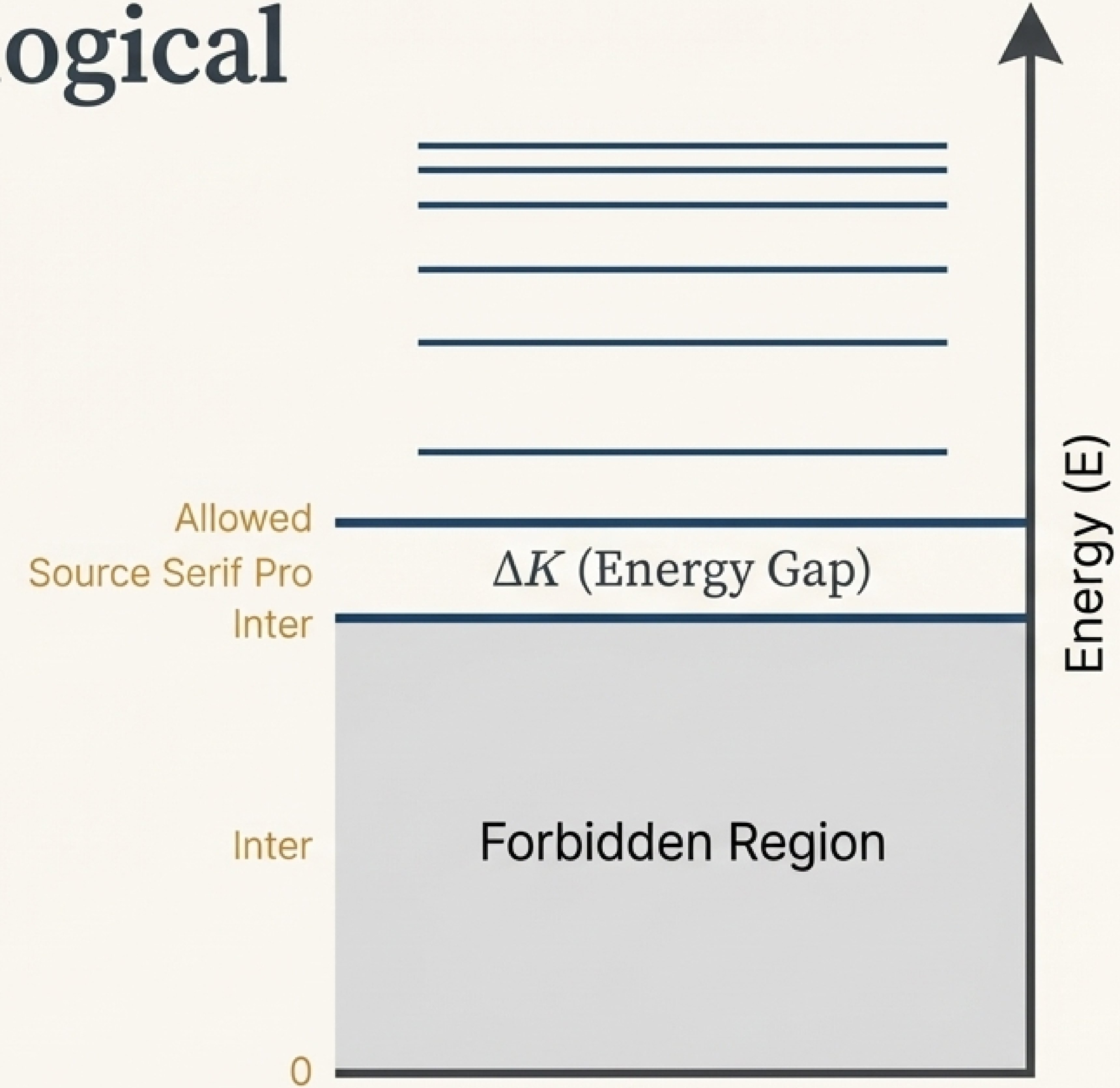
The Solution: A Topological Excitation Gap

Core Concept:

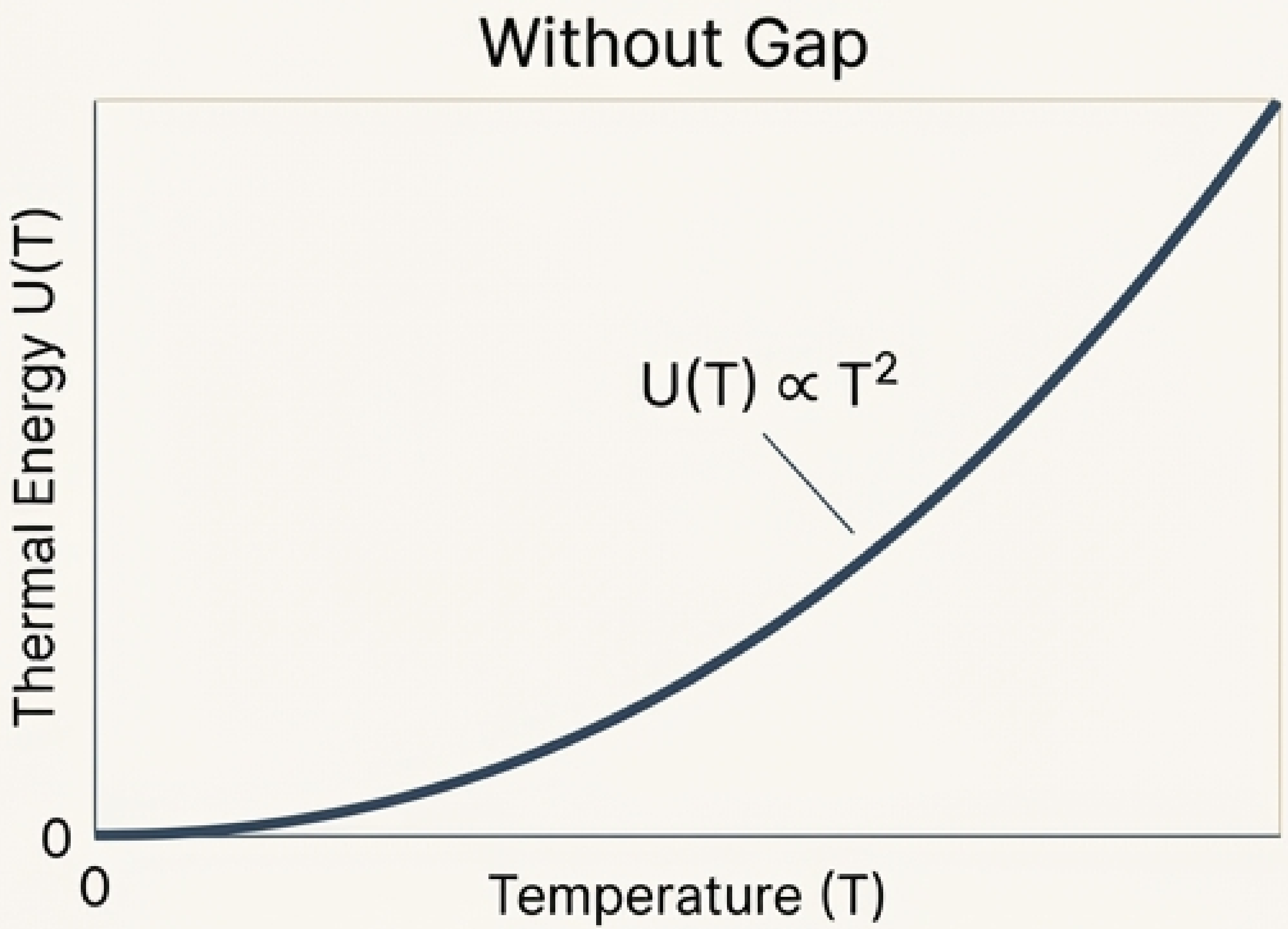
We propose a fundamental energy gap, ΔK. No Kelvin waves can be excited with less than this threshold energy. Higher modes must obey $E_{m,n} \geq \Delta K$.

Physical Motivation:

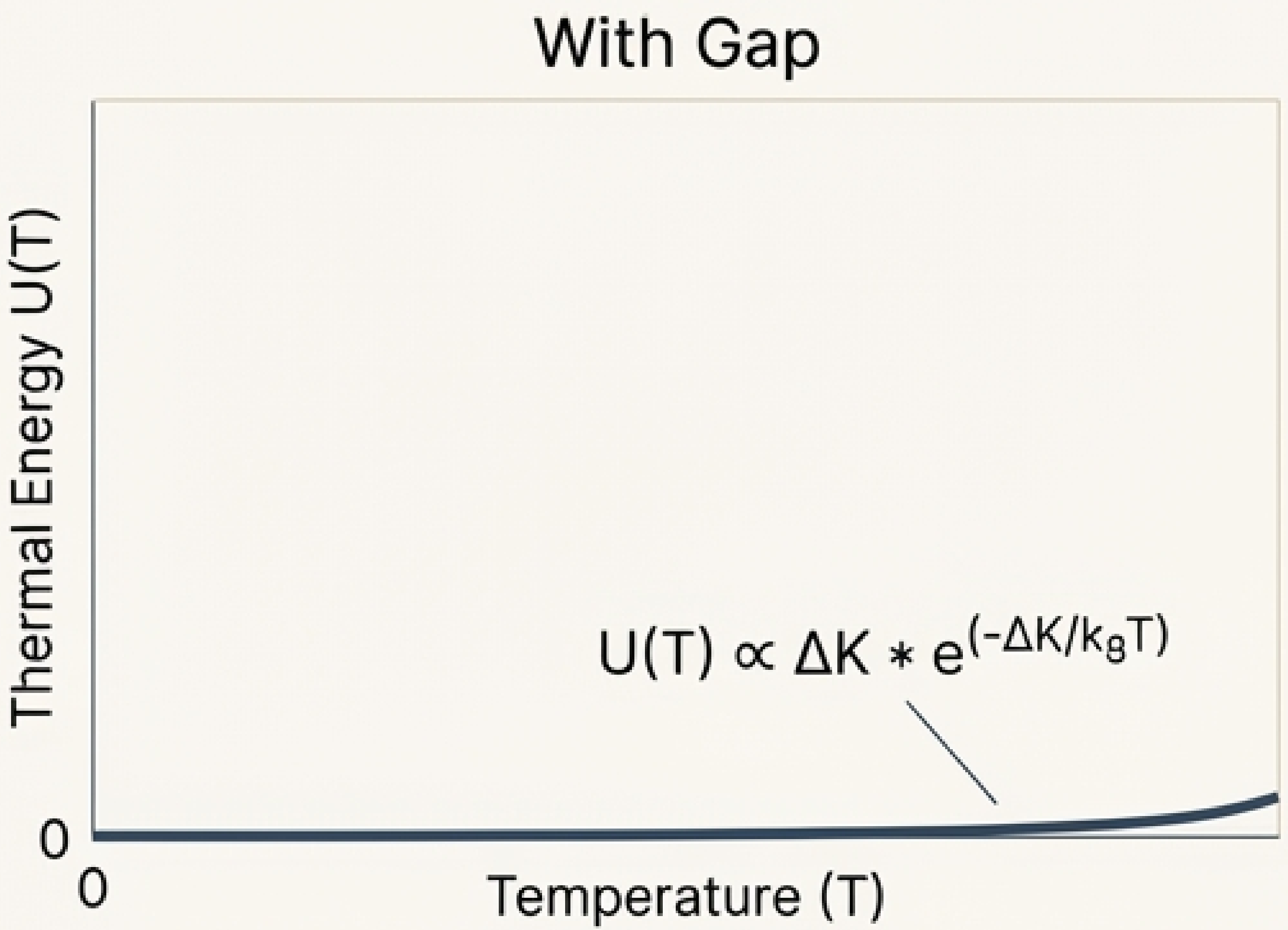
Knotted filaments can naturally produce such energy thresholds through reconnection constraints, curvature, and torsion.



The Freezing Effect: From Catastrophe to Quiescence



Thermal energy grows rapidly, causing catastrophic level shifts.



Thermal energy is exponentially suppressed, "freezing out" Kelvin modes.

The gap transforms the dangerous polynomial scaling of thermal energy into a negligible exponential tail at atomic temperatures, rendering the Kelvin modes inert.

NotebookLM

Quantifying the Solution: How Large Must the Gap Be?

Start with Reality in Source Serif Pro

Precision spectroscopy sets an incredibly stringent limit on energy level shifts.

$$a_{\mathrm{eff,n}} \lesssim 10^{-62}\ \mathrm{J\ K^{-2}}$$

Calculate Requirement in Source Serif Pro

To meet this limit, the thermal suppression factor must be immense.

$$\Delta K / k_8 T_{\mathrm{eff}} \gtrsim 60$$

Assume Conditions in Source Serif Pro

Using a conservative effective temperature bound from atomic physics.

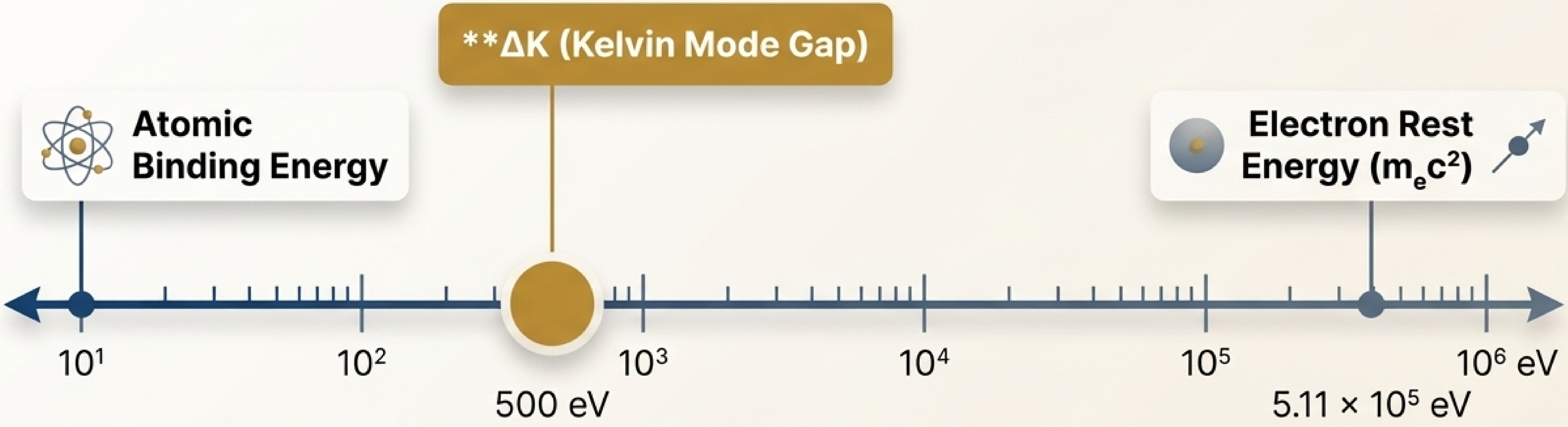
$$T_{\mathrm{eff}} \lesssim 10^5\ \mathrm{K}$$

Derive the Gap in Source Serif Pro

This implies a minimum required gap energy of:

$$\Delta K \gtrsim 500\ \mathrm{eV}$$

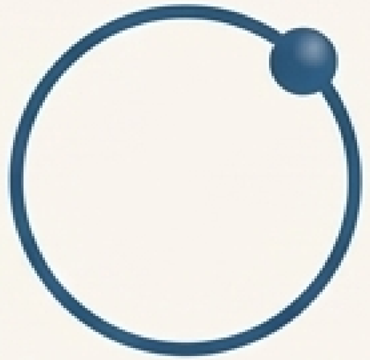
A Physically Plausible ‘Goldilocks’ Energy Scale



The gap is large enough to completely freeze Kelvin modes in ordinary atoms, yet small enough to be a plausible feature of the electron’s internal structure.

A Tale of Two Regimes: A Clean Separation of Scales

Low-Energy Regime (Kelvin-Frozen)



$$k_8 T \ll \Delta K$$

Kelvin modes are inert and suppressed. Hydrodynamic equilibrium of the whole filament dominates.

The model reproduces **Standard Quantum Mechanics**.

High-Energy Regime (Kelvin-Active)

$$\text{Energy} > \Delta K$$

Kelvin modes can be excited. The internal, vibrational dynamics of the filament become relevant.

The model predicts **New, Testable Phenomenology**

Vindicating Schrödinger: Deriving Quantum Mechanics from Hydrodynamics

With Kelvin modes suppressed, the stationary Schrödinger equation emerges a is an effective equation of state.

Step 1: The Functional

Start with the minimal free-energy functional for the vortex filament in the Kelvin-frozen regime:

$$F[\psi] = \int d^3r \left[\frac{\hbar^2}{2m_e} |\nabla \psi|^2 + V(r)|\psi|^2 \right]$$

Step 2: The Constraint

Minimizing this functional under normalization yields the Euler-Lagrange condition.

$$\int |\psi|^2 d^3r = 1$$

Step 3: The Result

The condition is the stationary Schrödinger equation:

$$\boxed{-\frac{\hbar^2}{2m_e} \nabla^2 \psi + V(r)\psi = E\psi}$$

The New Frontier: Where to Look for Kelvin Modes

Under what extreme conditions could the $\Delta K \approx 500$ eV gap be overcome, activating the filament's internal modes?

Ultra-High Acceleration

Probing effects in regimes with accelerations of order $a \sim 10^{22}\ \mathrm{m}/s^2$.

High-Energy Scattering

Searching for energy-loss features in keV-MeV scattering experiments.

Astrophysical & Cosmological Fluids

Investigating vortex reconnections in extreme cosmic environments.

A Consistent Model with Testable Predictions

1. The Problem

Ungapped Kelvin waves on vortex filaments predict thermal corrections to atomic spectra that are >20 orders of magnitude larger than observed, creating a thermodynamic catastrophe.

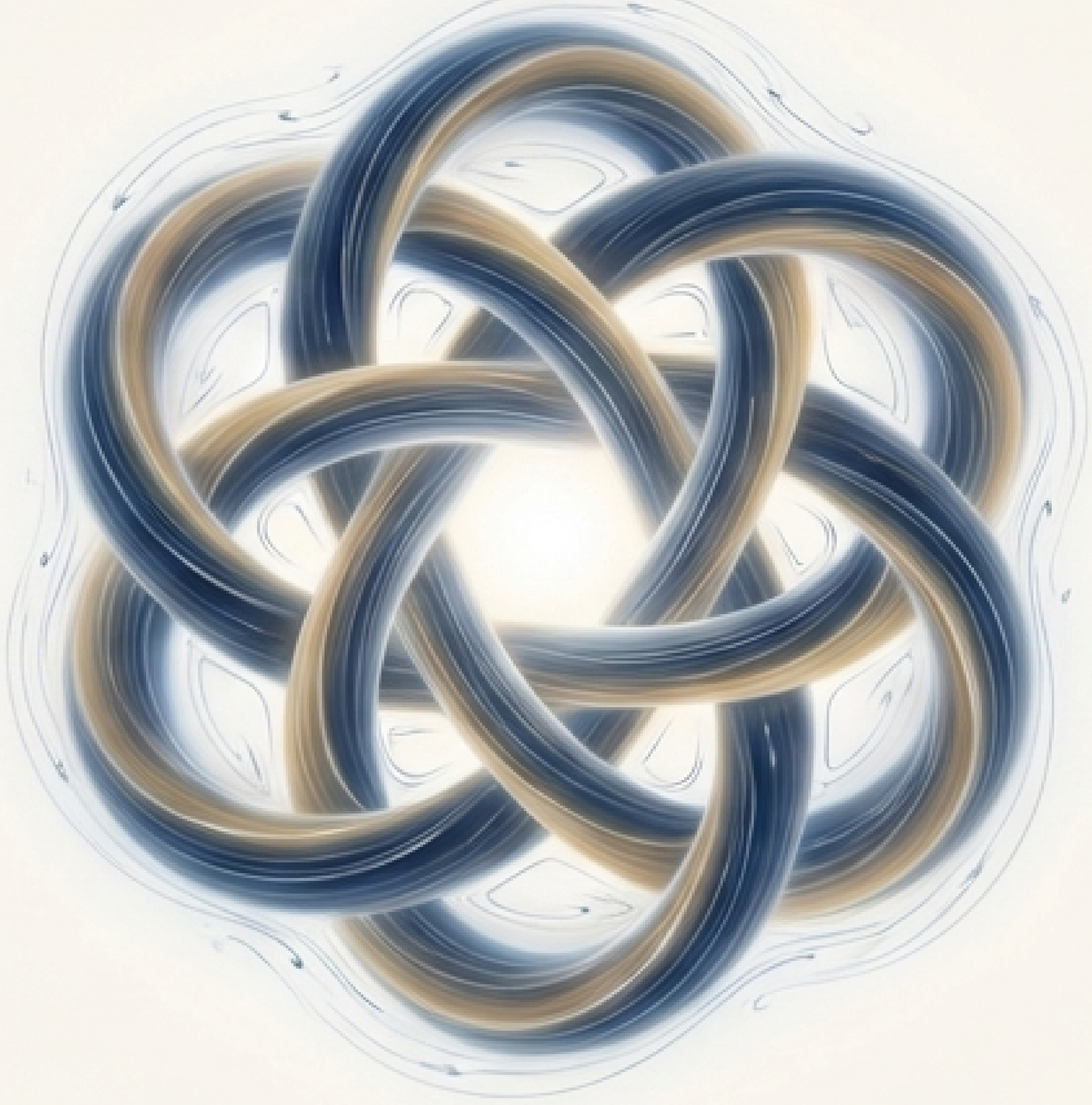
2. The Solution

A topological excitation gap of $\Delta K \gtrsim 500\ \mathrm{eV}$ exponentially suppresses these internal modes at atomic temperatures, ensuring the stability of the hydrogenic spectrum.

3. The Implication

This establishes a viable and consistent hydrodynamic model that recovers the Schrödinger equation at low energies and predicts new, testable physics in high-energy and high-acceleration regimes.

The Broader Picture: The Swirl-String Framework (SST)



Context

This gapped Kelvin mechanism provides a key consistency check for broader frameworks like SST, where electrons are modeled as topologically stable, knotted vortex filaments in a real, incompressible medium.

Function

It explains *why* low-energy physics is effectively quantum-mechanical while simultaneously predicting where internal degrees of freedom should become active.

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